

Midterm Study Guide

EC325: Econometrics | Colby College

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Unit 1: Introduction to Econometrics

What is Econometrics?

- Definition of econometrics and its emphasis on causal inference
- Examples of causal inference questions (health insurance and health, education and earnings, tariffs and production)
- Steps of an econometric project: model, data, estimate, interpret

Correlation vs. Causality

- Why correlation does not imply causation
- Selection bias (apples-to-oranges comparisons)
- Ceteris paribus condition: comparing apples-to-apples

Unit 2: Good Data Practices and R

Why Good Data Practices Matter

- The Reinhart-Rogoff example: spreadsheet errors led to influential but incorrect policy conclusions
- Errors were only discovered because a grad student attempted to replicate the results
- The replication standard: an independent researcher should be able to exactly duplicate results

Getting to Know Your Data

- Always examine your data before running any analysis
- Key questions: How many observations? Any missing data? Is the distribution reasonable? Any outliers or impossible values?
- Tools: descriptive statistics (`mean()`, `sd()`, `min()`, `max()`, `summary()`) and visualizations

Missing Data

- Use `is.na()` and `sum(is.na())` to detect missing values
- Use `na.rm = TRUE` to ignore missing values in calculations (e.g., `mean(x, na.rm = TRUE)`)
- Data can be missing at random or missing systematically — the latter can bias results

R Basics

- **Objects and assignment:** Store values using `<-` (e.g., `x <- 3`)
- **Vectors:** Combine values with `c()` (e.g., `ages <- c(19, 20, 21)`)
- **Functions:** Pre-built operations that take arguments (e.g., `mean(x)`, `seq(from = 1, to = 10, by = 2)`)
- **Data frames:** Rows = observations, columns = variables; access columns with `$` (e.g., `mtcars$mpg`)
- **Data types:** numeric, character, logical, factor — functions behave differently depending on type
- **Packages:** Install once with `install.packages()`, load every session with `library()`

The Tidyverse and the Pipe

- The **tidyverse** is a collection of R packages for data science (`dplyr`, `ggplot2`, `readr`, etc.)
- The pipe `|>` passes the result on its left as the first argument to the function on its right — read as “and then”

Key dplyr Functions

You should know what each of these functions does and be able to read/write basic code using them:

- **`filter()`:** Subset rows by a condition (e.g., `filter(mpg > 25)`)
- **`select()`:** Choose or drop specific columns (e.g., `select(mpg, cyl, wt)`)

- **mutate():** Create new variables or transform existing ones (e.g., `mutate(kmpl = mpg * 0.425)`)
- **summarize():** Collapse data into summary statistics (e.g., `summarize(avg = mean(mpg))`)
- **group_by():** Group data before summarizing — pair with `summarize()` for grouped statistics
- **arrange():** Sort rows by a column; use `desc()` for descending order
- **count():** Count rows per unique value — shortcut for `group_by() |> summarize(n = n())`
- **Chaining:** Multiple verbs can be piped together to build complex operations step-by-step

Descriptive Statistics Functions

- `mean()`, `sd()`, `median()`, `min()`, `max()`: operate on vectors (columns)
- `n()`: counts observations (used inside `summarize()`)
- `summary()`: quick overview of min, quartiles, mean, max for all columns
- `glimpse()`: shows variable names, types, and first few values

Visualization with ggplot2

- Plots are built layer by layer: data → geometry → aesthetics → labels
- `ggplot(data)`: initialize the plot
- `aes(x, y, color)`: map variables to visual properties
- Key geometries: `geom_point()` (scatterplot), `geom_histogram()` (distribution), `geom_smooth(method = "lm")` (line of best fit), `geom_col()` (bar chart)
- `labs()`: add title, axis labels, legend titles
- `theme_minimal()`: clean visual theme

Good Coding Practices

- Always work in scripts (.R files), not the console
- Use R Projects (.Rproj) to set your working directory automatically
- Comment your code with `#` to explain what it does and why
- Use descriptive variable names (e.g., `avg_wage_by_state` not `x1`)

Unit 3: RCTs and the Potential Outcomes Framework

Randomized Control Trials

- How RCTs work: random assignment to treatment and control groups
- Why randomization eliminates selection bias
- The Law of Large Numbers (LLN) and its role in ensuring balanced groups

The Potential Outcomes Framework

- Notation: Y_{1i} (treated outcome), Y_{0i} (untreated outcome)
- Causal effect: $\kappa = Y_{1i} - Y_{0i}$
- The fundamental problem of causal inference: we only observe one potential outcome per unit
- Decomposition of the simple difference in means into a causal effect and a selection bias term:

$$E[Y_i|D_i = 1] - E[Y_i|D_i = 0] = \kappa + \underbrace{(E[Y_{0i}|D_i = 1] - E[Y_{0i}|D_i = 0])}_{\text{Selection Bias}}$$

- How randomization sets the selection bias term to zero
- The Oregon Health Insurance Experiment as an applied example

Unit 4: Bivariate OLS

Conditional Expectations

- Definition and interpretation of $E[Y|X = x]$
- Conditional expectations as the foundation of regression

The Population Regression Model

- Formula: $y = \beta_0 + \beta_1 x + \mu$
- Definitions: dependent variable (y), independent variable (x), error term (μ), population parameters (β_0, β_1)
- Distinction between population parameters and sample estimates

Deriving the OLS Estimates

- OLS minimizes the Sum of Squared Residuals (SSR): $SSR = \sum_{i=1}^n (y_i - \hat{y}_i)^2$
- Slope formula: $\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{Cov(x, y)}{Var(x)}$
- Intercept formula: $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$
- Relationship between $\hat{\beta}_1$ and correlation: $\hat{\beta}_1 = \rho_{xy} \frac{\sigma_y}{\sigma_x}$

Interpreting OLS Coefficients

- $\hat{\beta}_0$: Predicted y when $x = 0$
- $\hat{\beta}_1$: Marginal effect — the change in $\hat{y}$ for a one-unit increase in x

Goodness-of-Fit

- Total Sum of Squares (SST): $\sum (y_i - \bar{y})^2$
- Explained Sum of Squares (SSE): $\sum (\hat{y}_i - \bar{y})^2$
- $R^2 = SSE/SST = 1 - SSR/SST$

Statistical Assumptions for Unbiasedness (Bivariate)

- **GM 1 (Linearity in Parameters)**: The model is linear in β — non-linear transformations of data are allowed
- **GM 2 (Random Sampling)**: The sample is randomly drawn from the population
- **GM 3 (No Perfect Multicollinearity)**: The independent variable is not constant
- **GM 4 (Zero Conditional Mean)**: $E[\mu|x] = 0$ — the error is uncorrelated with x
- If GM 1–4 hold, OLS is **unbiased**: $E[\hat{\beta}_j] = \beta_j$

Unit 5: Multivariate OLS

Limits of Bivariate OLS

- When bivariate OLS does and does not generate ceteris paribus estimates
- Susceptibility to omitted variable bias

The Multivariate OLS Model

- General formula: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + \mu$
- Interpretation: $\hat{\beta}_j$ is the ceteris paribus effect of x_j on y , holding all other included variables constant
- Zero Conditional Mean with k variables: $E[\mu|x_1, x_2, \dots, x_k] = 0$
- Multivariate OLS as a matching estimator

Omitted Variable Bias (OVB)

- **Two conditions for OVB:** (1) the omitted variable affects y , AND (2) the omitted variable is correlated with an included x
- **OVB formula:** $E[\tilde{\beta}_1] = \beta_1 + \beta_2 \cdot \delta_1$, where $Bias = \beta_2 \cdot \delta_1$
- β_2 : effect of the omitted variable on y
- δ_1 : correlation between the omitted variable and the included variable
- **Direction of bias:** determined by the signs of β_2 and δ_1

$Corr(x_1, x_2)$	$\beta_2 > 0$	$\beta_2 < 0$
$\delta_1 > 0$	Positive Bias	Negative Bias
$\delta_1 < 0$	Negative Bias	Positive Bias

Goodness-of-Fit in Multivariate Models

- R^2 mechanically increases with more variables
- Adjusted R^2 ($\bar{R}^2$) penalizes for additional parameters; always $\leq R^2$
- $\bar{R}^2$ is preferred in multivariate models

Gauss-Markov Assumptions (Multivariate)

- GM 1–4 as applied to the multivariate setting
- GM 3 (No Perfect Multicollinearity): no exact linear relationships among the x variables

GM Assumption 5 and Properties of the OLS Estimator

GM Assumption 5: Homoskedasticity

- $Var(\mu|x_1, \dots, x_k) = \sigma^2$ — constant error variance
- **Heteroskedasticity:** when error variance depends on the x values
- If GM 1–5 hold, OLS is **BLUE** (Best Linear Unbiased Estimator)

Variance of the OLS Estimator

- Formula: $Var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1-R_j^2)}$
- Three drivers of variance:
 - σ^2 (error variance): more noise $\rightarrow$ higher variance
 - SST_j (sample variation in x_j): more variation $\rightarrow$ lower variance
 - R_j^2 (multicollinearity): more correlation among x 's $\rightarrow$ higher variance
- Larger sample size increases SST_j , reducing variance

Estimating σ^2 from the Residuals

The variance formula contains σ^2 , a population parameter we cannot observe. We need to estimate it from sample data. The logic proceeds as follows:

1. From homoskedasticity and zero conditional mean: $Var(\mu_i) = \sigma^2 = E[\mu_i^2]$
2. So to estimate σ^2 , we need a good estimate of the error terms μ_i
3. We cannot observe the true errors μ_i , but we can compute the **residuals**: $\hat{\mu}_i = y_i - \hat{y}_i$
4. The average squared residual (adjusted for degrees of freedom) is an unbiased estimator of σ^2 :

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n \hat{\mu}_i^2}{n - k - 1} = \frac{SSR}{n - k - 1}$$

- We divide by $n - k - 1$ (not n) because estimating $k + 1$ parameters uses up $k + 1$ degrees of freedom
- In the bivariate case ($k = 1$), this is $n - 2$

The Standard Error of $\hat{\beta}_j$

- Plugging $\hat{\sigma}^2$ into the variance formula and taking the square root gives the standard error:

$$se(\hat{\beta}_j) = \frac{\hat{\sigma}}{\sqrt{SST_j(1 - R_j^2)}}$$

- The standard error measures the precision of our estimate — it is the estimated standard deviation of the sampling distribution of $\hat{\beta}_j$
- **Key chain of logic:** We go from a formula involving an unknown population parameter (σ^2) to a computable standard error by replacing σ^2 with its sample estimate from the residuals

The Bias/Variance Tradeoff

- Including more controls: reduces bias but can inflate variance
- Omitting controls: lower variance but introduces bias
- Larger samples reduce variance, making the tradeoff less severe

Unit 6: Statistical Inference (through P-Values)

The Sampling Distribution

- To do inference, we need three things about $\hat{\beta}_j$: its **center**, **spread**, and **shape**
- From Gauss-Markov we already know the center (unbiasedness: $E[\hat{\beta}_j] = \beta_j$) and spread (standard error)
- The missing piece is the shape — which is what the normality assumption and the CLT provide

The Normality Assumption (MLR.6)

- $\mu \mid x_1, \dots, x_k \sim Normal(0, \sigma^2)$
- Conditional on the x 's, the error term follows a bell curve centered at zero
- Implies: $\hat{\beta}_j \sim Normal[\beta_j, Var(\hat{\beta}_j)]$
- Combined with Gauss-Markov, this gives us the full sampling distribution of $\hat{\beta}_j$

The Central Limit Theorem (CLT)

- Even if errors are **not** normally distributed, the CLT tells us that $\hat{\beta}_j$ is **approximately** normal in large samples:

$$\frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)} \xrightarrow{d} Normal(0, 1) \quad \text{as } n \rightarrow \infty$$

- This works because $\hat{\beta}_j$ is a weighted average of the y_i 's, and averages become normal
- **Asymptotic normality**: we don't need the normality assumption to hold exactly — just a large enough n
- This is important in practice because many economic variables are skewed (wages, health spending, etc.)

Hypothesis Testing

- **Null hypothesis:** $H_0 : \beta_j = 0$ (no effect)
- **Alternative hypothesis:** $H_A : \beta_j \neq 0$
- **Logic:** Assume H_0 is true, check whether the observed estimate is likely under this assumption
- **Key insight:** Under H_0 , we know the entire sampling distribution of $\hat{\beta}_j$ — it is centered at zero (unbiasedness + null), has a known spread (SE), and is bell-shaped (CLT/CLM). We know what “no effect” looks like.

The T-Statistic

- $t_{\hat{\beta}_j} = \frac{\hat{\beta}_j}{se(\hat{\beta}_j)}$
- Measures how many standard errors the estimate is from zero
- **Rule of thumb (large samples):** $|t| > 2$ implies significance at the 5% level

Type I Error and Significance Levels

- **Type I error** (“false positive”): rejecting H_0 when it’s actually true
- **Significance level** (α): our tolerance for this mistake (commonly 10%, 5%, 1%)
- **Critical value** (c_α): the t -value that defines the rejection region
 - For large samples: $c_{0.10} \approx 1.645$, $c_{0.05} \approx 1.96$, $c_{0.01} \approx 2.576$
- **Decision rule:** Reject H_0 if $|t| > c_\alpha$

P-Values

- The smallest significance level at which H_0 would be rejected
- Interpretation: the probability of observing a t -statistic as extreme as ours, if H_0 were true
- **Not** the probability that H_0 is true
- Rejection rule: if p-value $< \alpha$, reject H_0

Reading a Regression Table

- **Rows:** Each row corresponds to one regressor (x_j); the top number is $\hat{\beta}_j$ and the number in parentheses below is $se(\hat{\beta}_j)$

- **Columns:** Each column is a different regression specification (e.g., bivariate, then progressively adding controls); placing them side-by-side shows how coefficients change as controls are added
- **Stars:** Indicate statistical significance, but the thresholds vary — always check the notes at the bottom for the convention
- **Bottom rows:** Summary statistics such as N , R^2 , and adjusted R^2
- **Common mistakes:** The parentheses contain standard errors (not t -statistics, not confidence intervals); stars tell you about statistical significance but not practical importance